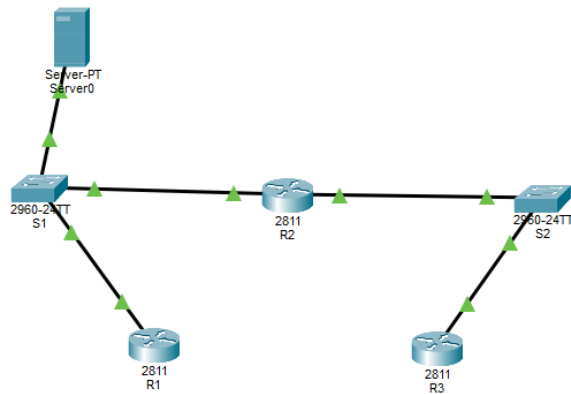


Week 3 Practical

Topology



DNS configuration

R1

```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip name-server 10.10.10.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config | include name-server
ip name-server 10.10.10.1
Router#
```

R2

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface F0/0
Router(config-if)#ip name-server 10.10.10.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config | include name-server
^
% Invalid input detected at '^' marker.

Router#
Router#show running-config | include name-server
ip name-server 10.10.10.2
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface F0/1
Router(config-if)#ip name-server 10.10.20.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config | include name-server
ip name-server 10.10.20.2
Router#
```

R3

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip name-server 10.10.20.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show running-config | include name-server
ip name-server 10.10.20.1
Router#
```

DNS Ping

```
Router>enable
Router#ping server.local
Translating "server.local"...domain server (10.10.10.1)|
```

Show arp

```
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 0001.6456.C501 ARPA FastEthernet0/0
```

Clear arp-cache and ping

```
Router#clear arp-cache
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 5/23/36 ms
```

Reflection Questions

20. An ARP entry gives the router the MAC address associated with a known IP address, while a routing table tells the router where to forward the packet at the IP/routing level but does not provide the destination MAC address.

21. A routing table determines the next-hop IP address and outgoing interface, but Ethernet networks require a destination MAC address to actually deliver the frame.